

Arrays-Order to the Universe

The types of data structures we have been using are (

Lists-We have worked with these and they are nice because they are mutable and easy to work with. But they are not great with numbers.

Tuples-We are not spending time on them. They are like lists with a funny name but they are immutable. So they can be good if you need something not to change. But we don't use them because they don't do much with numbers.

Dictionaries-We use these a little. They are thought of key:value pairs and are created with {}. We have used these already when defining our "props" on the graphs. It lets you pass a few keyword values at once. We won't use these much but you will come across them.

Numpy arrays-we have started using these. We have seen they are easy to plot and to do math with. But they are not great with large datasets with lots of different columns and with missing data. But they are the basis for a lot of things in python so you always build off of numpy.

Pandas Dataframes-We have started these. these are like supercharged numpy arrays that give you a lot more information. If you could imagine that you could name the rows and columns in a numpy array it starts to get you there. Sort of like an excel sheet in the computer memory but more powerful. Plus they are good with dates and re-ordering. So these are good for complex datasets where we want to name variables. We will mainly be using these and building everything off of them.

But how do we think about data?

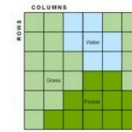
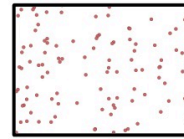
When we usually think about data we think about tabular data. This is an excel sheet. Just a table of the data we have. this is the dataframe we read in. But there are really three data types we might run into

- Tabular Data
- Vector Data
- Raster Data

The picture below explain them

Data types

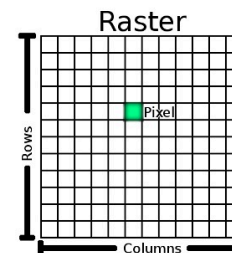
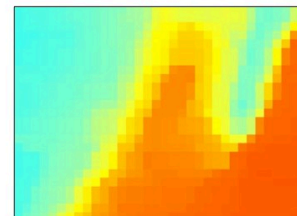
- Vector
 - Points
 - Lines
 - Polygons
- Raster
 - Digital Elevation Models (DEM)
 - Ortho imagery (aerial photography)
 - Satellite imagery
- Tabular data
 - Attributes
 - Databases



ID	Name	APRIS	STATE ABBR	STATE FIPS	SUMMARY
1	Alaska	AK	Alaska	02	AK
2	Alabama	AL	Alabama	01	AL
3	Arizona	AZ	Arizona	04	AZ
4	Arkansas	AR	Arkansas	05	AR
5	California	CA	California	06	CA
6	Colorado	CO	Colorado	08	CO
7	Connecticut	CT	Connecticut	09	CT
8	Delaware	DE	Delaware	10	DE
9	District of Columbia	DC	District of Columbia	11	DC
10	Florida	FL	Florida	12	FL
11	Georgia	GA	Georgia	13	GA
12	Hawaii	HI	Hawaii	15	HI
13	Idaho	ID	Idaho	16	ID
14	Illinois	IL	Illinois	17	IL
15	Indiana	IN	Indiana	18	IN
16	Iowa	IA	Iowa	19	IA
17	Kansas	KS	Kansas	20	KS
18	Kentucky	KY	Kentucky	21	KY
19	Louisiana	LA	Louisiana	22	LA
20	Maine	ME	Maine	23	ME
21	Maryland	MD	Maryland	24	MD
22	Massachusetts	MA	Massachusetts	25	MA
23	Michigan	MI	Michigan	26	MI
24	Minnesota	MN	Minnesota	27	MN
25	Mississippi	MS	Mississippi	28	MS
26	Missouri	MO	Missouri	29	MO
27	Montana	MT	Montana	30	MT
28	Nebraska	NE	Nebraska	31	NE
29	Nevada	NV	Nevada	32	NV
30	New Hampshire	NH	New Hampshire	33	NH
31	New Jersey	NJ	New Jersey	34	NJ
32	New Mexico	NM	New Mexico	35	NM
33	New York	NY	New York	36	NY
34	North Carolina	NC	North Carolina	37	NC
35	North Dakota	ND	North Dakota	38	ND
36	Ohio	OH	Ohio	39	OH
37	Oklahoma	OK	Oklahoma	40	OK
38	Oregon	OR	Oregon	41	OR
39	Pennsylvania	PA	Pennsylvania	42	PA
40	Rhode Island	RI	Rhode Island	43	RI
41	South Carolina	SC	South Carolina	44	SC
42	South Dakota	SD	South Dakota	45	SD
43	Tennessee	TN	Tennessee	46	TN
44	Texas	TX	Texas	48	TX
45	Utah	UT	Utah	49	UT
46	Vermont	VT	Vermont	50	VT
47	Virginia	VA	Virginia	51	VA
48	Washington	WA	Washington	52	WA
49	West Virginia	WV	West Virginia	53	WV
50	Wisconsin	WI	Wisconsin	54	WI
51	Wyoming	WY	Wyoming	55	WY

Raster data

- Areas broken into “pixels” or cells
- Each cell contains data
- Good at representing dense data:
 - Land cover
 - Elevation



Vector data are used with GIS. They are for putting lines, points, or polygons on a map. For example putting a shoreline on a map or a road or a lake. We are not going to use them much this semester.

But first today we are going to talk about Raster data and two dimensional arrays. Raster data is really a 2d array.

Today lets just work with two dimensional numpy arrays. You can have arrays of as many dimensions as you want but I have trouble comprehending at three and more dimensions.

```
In [12]: %matplotlib ipynb
import matplotlib.pyplot as plt
import numpy as np
```

```
from scipy import stats
import pandas as pd
np.set_printoptions(legacy='1.25') #removes the funny printing
```

```
In [2]: oneD=np.array([1,2,3,4,5,6,7,8,9,10]) # does anyone remember that band?
```

```
In [3]: oneD
```

```
Out[3]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

```
In [4]: #review-what will this print?
#oneD[0:10:2]
```

```
In [4]: twoD=np.array([[1,2,3,4],[5,6,7,8]])
```

```
In [5]: twoD
```

```
Out[5]: array([[1, 2, 3, 4],
               [5, 6, 7, 8]])
```

Do you see what I just did? It is 2 one dimensional arrays together to make a 2-dimensional array or table!

```
In [6]: type(twoD)
```

```
Out[6]: numpy.ndarray
```

```
In [7]: len(twoD)
```

```
Out[7]: 2
```

```
In [8]: twoD.shape
```

```
Out[8]: (2, 4)
```

```
In [9]: np.shape(twoD)
```

```
Out[9]: (2, 4)
```

```
In [10]: twoD.size
```

```
Out[10]: 8
```

You can see what you can do with the np array by typing twoD. and then tab and you see the functions available. try one

```
In [108... twoD.
```

Now lets try slicing. Remember it is rows and then columns. See if you can guess before uncommenting and running. This picture is just to help you and you don't need to load it.

```
In [77]: from IPython.display import Image
Image(filename='array-axes.png',width=400)
```

Out [77]:

		axis 1		
		0	1	2
axis 0	0	0,0	0,1	0,2
	1	1,0	1,1	1,2
	2	2,0	2,1	2,2

In [13]: `twoD[0,0]`

Out[13]: 1

In [14]: `twoD[1,0]`

Out[14]: 5

In [15]: `twoD[:,:]`Out[15]: `array([[1, 2, 3, 4],
[5, 6, 7, 8]])`In [16]: `#twoD[:,0]`In [111... `#twoD[:,1]`In [112... `#twoD[0,:]`In [113... `#twoD[1,:]`In [114... `#twoD[0,0]`In [115... `#twoD[3,3]`In [116... `#twoD[1,3]`

`np.vstack` adds a row. So lets make our array bigger and keep going!

In [16]: `twoD=np.vstack((twoD,[9,10,11,12]))`In [17]: `twoD`

```
Out[17]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12]])
```

```
In [18]: twoD=np.vstack((twoD,[13,14,15,16]))
```

```
In [19]: twoD
```

```
Out[19]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12],
               [13, 14, 15, 16]])
```

Now lets do some more slicing!

remember. For numpy it is

[start:stop:skip]

if you list multiple items and leaving one out assumes the last one is missing

so that means [1::] is one to the end by 1.

If you have a 2d array it will be [start:stop:skip,start:stop:skip] for the rows and then the columns

```
In [20]: twoD[:,:]
```

```
Out[20]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12],
               [13, 14, 15, 16]])
```

```
In [17]: #twoD[::2,::2]
```

```
In [18]: #twoD[2:,2:]
```

```
In [19]: #twoD[1:3,1:3]
```

```
In [20]: #twoD[1::2,:]
```

Now you can set the numbers in different places.

```
In [21]: twoD[3,3]=100
```

```
In [22]: twoD
```

```
Out[22]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12],
               [13, 14, 15, 100]])
```

```
In [23]: twoD[1:3,1:3]=55
```

```
In [24]: twoD
```

```
Out[24]: array([[ 1,  2,  3,  4],
               [ 5, 55, 55,  8],
               [ 9, 55, 55, 12],
               [13, 14, 15, 100]])
```

you can use a function to set numbers!

```
In [25]: twoD[0,:]=np.arange(10,14)
```

```
In [26]: twoD
```

```
Out[26]: array([[10, 11, 12, 13],
               [ 5, 55, 55,  8],
               [ 9, 55, 55, 12],
               [13, 14, 15, 100]])
```

you can reshape the array if you want to.

```
In [27]: print (twoD.reshape(16,1))
```

```
[[ 10]
 [ 11]
 [ 12]
 [ 13]
 [  5]
 [ 55]
 [ 55]
 [  8]
 [  9]
 [ 55]
 [ 55]
 [ 12]
 [ 13]
 [ 14]
 [ 15]
 [100]]
```

```
In [28]: print (np.reshape(twoD,(1,16)))
```

```
[[ 10  11  12  13   5  55  55   8   9  55  55  12  13  14  15 100]]
```

```
In [29]: print (twoD.reshape(8,2))
```

```
[[ 10  11]
 [ 12  13]
 [  5  55]
 [ 55   8]
 [  9  55]
 [ 55  12]
 [ 13  14]
 [ 15 100]]
```

```
In [30]: print (twoD)
```

```
[[ 10  11  12  13]
 [  5  55  55   8]
 [  9  55  55  12]
 [ 13  14  15 100]]
```

The shape is back to how we had it because we never changed because we only printed it. we never set it.

Now this is where we intersect with Raster Data

Add Color!

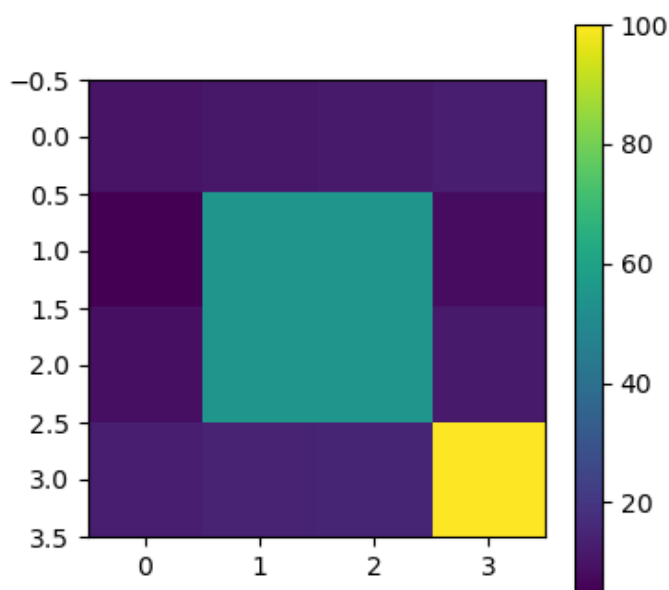
You can visualize the whole array! This just colors the array/grid we have by its values

This is raster data. It is like satellite data.

```
In [31]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
cax=ax.imshow(twoD)
fig.colorbar(cax)
```

```
Out[31]: <matplotlib.colorbar.Colorbar at 0x145a93e90>
```

Figure



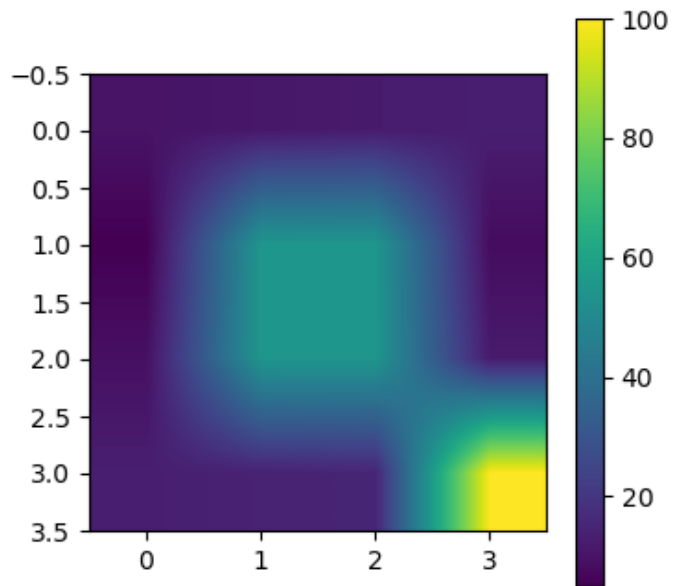
If you want to make the edges of each box smooth we need to interpolate the data. I chose interpolation='bilinear' but there are many options

https://matplotlib.org/gallery/images_contours_and_fields/interpolation_methods.html

```
In [32]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
cax=ax.imshow(twoD,interpolation='bilinear')
fig.colorbar(cax)
```

```
Out[32]: <matplotlib.colorbar.Colorbar at 0x14794bcd0>
```

Figure

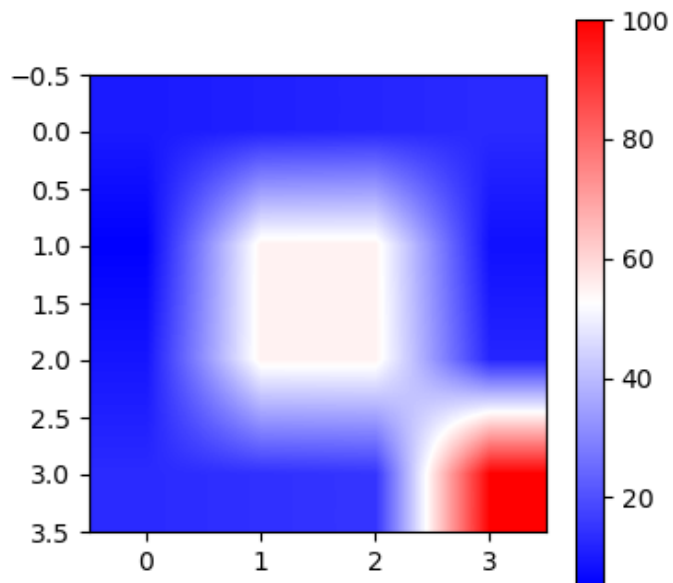


We can change the colorbar. This is another keyword argument

```
In [33]: fig,ax=plt.subplots()  
fig.set_size_inches(4,4)  
cax=ax.imshow(twoD,interpolation='bilinear',cmap='bwr')  
fig.colorbar(cax)
```

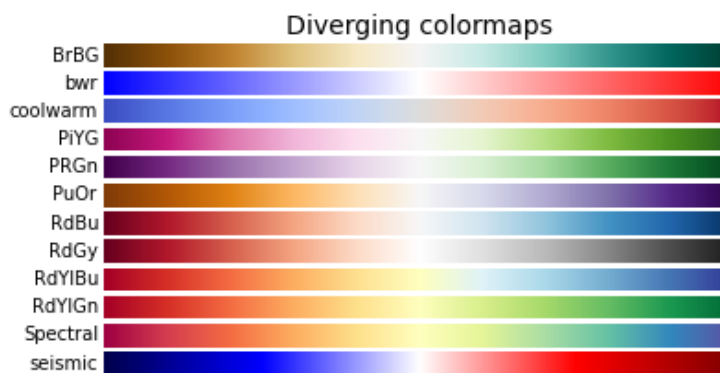
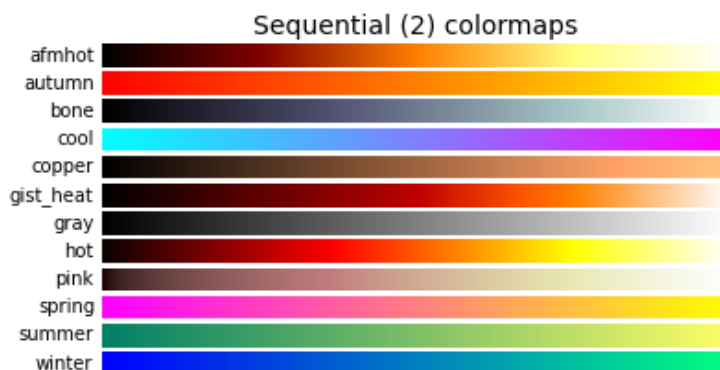
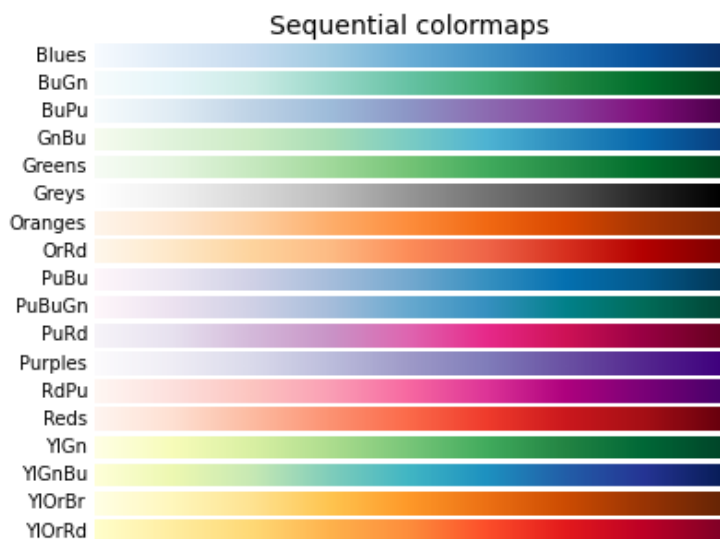
Out[33]: <matplotlib.colorbar.Colorbar at 0x147a6a290>

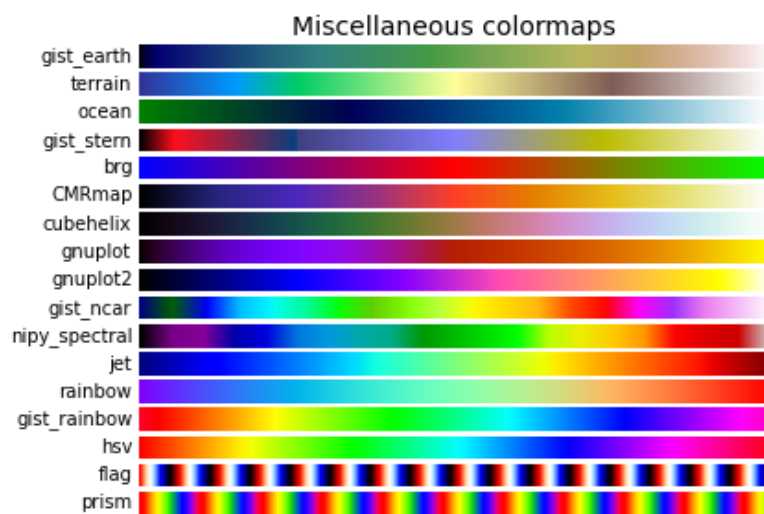
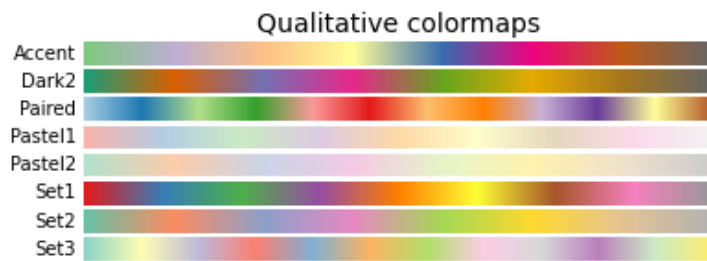
Figure



Now make your own. Here is just one [list](#). You can google colormaps python.

In [140... *#See below for the colormap code. I also just ran the code off the web to make them*



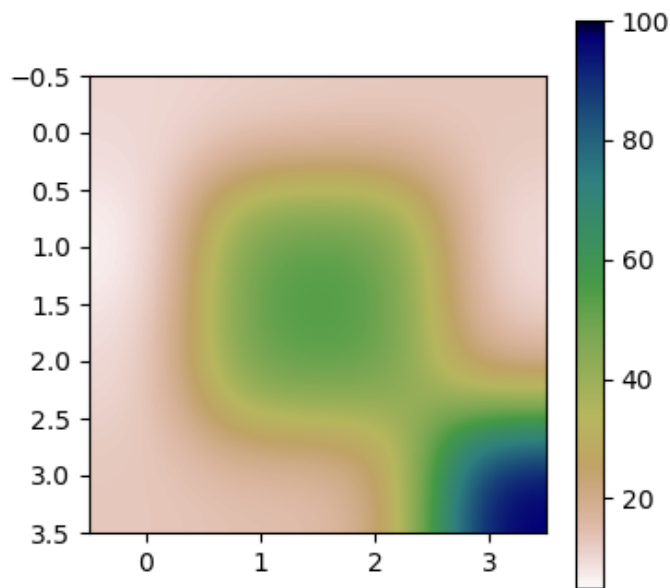


- Plot the same plot with a color bar of your choice.
- reverse the color bar by adding `_r` to the name.

```
In [34]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
cax=ax.imshow(twoD,cmap='gist_earth_r',interpolation='gaussian')
fig.colorbar(cax)
```

```
Out[34]: <matplotlib.colorbar.Colorbar at 0x1479a2290>
```

Figure



Now we are going to intersect back with pandas and dataframes.

We are going to

- read in a csv using pandas.
- this gives us a big dataframe which we can convert to a 2 dimensional array that is 100x100 in size
- then we can plot/map it.

https://github.com/bmaillou/BigDataPython/blob/master/my_first_csv.csv

```
In [35]: df=pd.read_csv('my_first_csv.csv',header=None)
```

```
In [36]: twoD=df.values # this takes the dataframe and turns it into an array
```

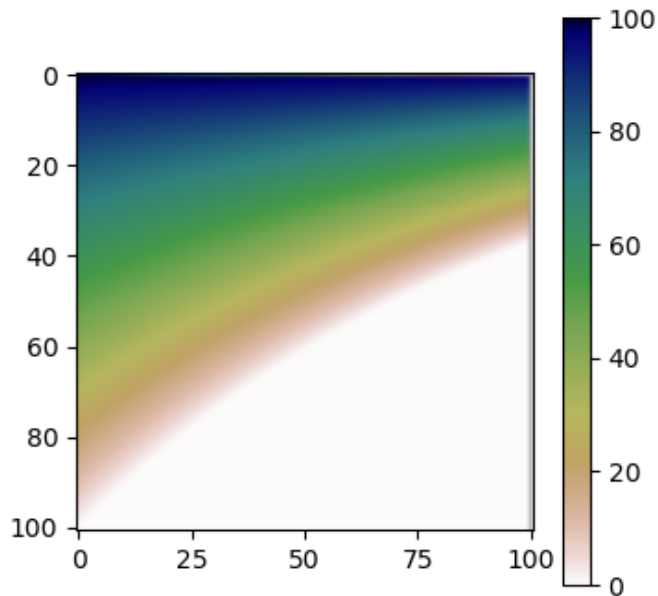
```
In [37]: print(twoD)
```

```
[ [1.00000000e+02 9.90000000e+01 9.80000000e+01 ... 2.00000000e+00
  1.00000000e+00 0.00000000e+00]
 [9.90000000e+01 9.89900000e+01 9.89799000e+01 ... 9.73484817e+01
  9.73219665e+01 nan]
 [9.80000000e+01 9.79800000e+01 9.79598000e+01 ... 9.46969634e+01
  9.46439330e+01 nan]
 ...
 [2.00000000e+00 1.02000000e+00 3.02000000e-02 ... 0.00000000e+00
  0.00000000e+00 nan]
 [1.00000000e+00 0.00000000e+00 0.00000000e+00 ... 0.00000000e+00
  0.00000000e+00 nan]
 [0.00000000e+00 0.00000000e+00 0.00000000e+00 ... 0.00000000e+00
  0.00000000e+00 nan]]
```

```
In [38]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
cax=ax.imshow(twoD,cmap='gist_earth_r',interpolation='gaussian')
fig.colorbar(cax)
```

Out[38]: <matplotlib.colorbar.Colorbar at 0x147c368d0>

Figure



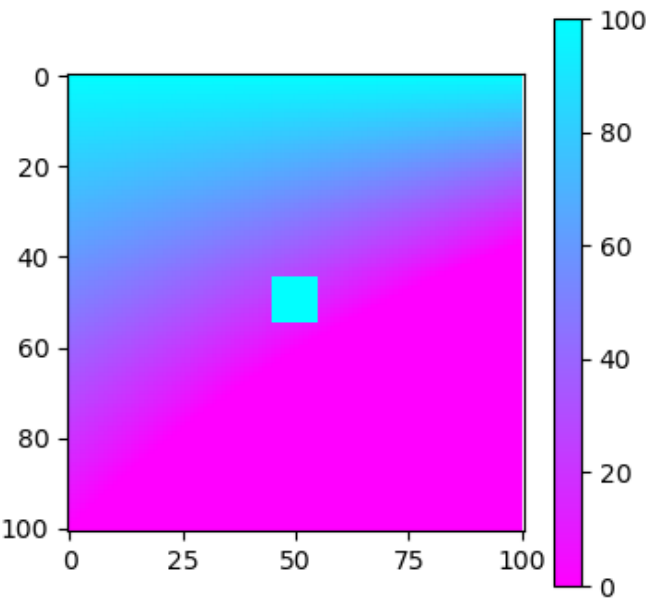
Remember we can alter the array. I am going to add a square in the middle of value 100...

```
In [39]: twoD[45:55,45:55]=100
```

```
In [40]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
cax=ax.imshow(twoD,cmap='cool_r',interpolation='none')
fig.colorbar(cax)
```

Out[40]: <matplotlib.colorbar.Colorbar at 0x147c72350>

Figure



```
In [ ]:
```

Now you need to Read in Brian.csv.

It is a 2d array. Plot it with imshow and tell me what it looks like. you will be shocked at what it shows.....

<https://github.com/bmaillou/BigDataPython/blob/master/Brian.csv>

```
In [ ]:
```

Homework hint: Think about how the data from Brian.csv is stored and how you can change it....

Now back to Tabular data.

For most of our data we usually work with tabular data. One example is if the first column is X and the remaining columns are all differnt Y's. Here is an example. Read in the oneX_manyY.csv file. print it to see it. Then plot all the values. This is really how we think of excel. But we give the columns nicer names

X Value	First Y Value	Second Y Value	Third Y Value	Fourth Y Value
1.0	1.0	10.0	4.0	6.0
2.0	2.0	9.0	4.0	6.0
3.0	3.0	8.0	4.0	6.0
4.0	4.0	7.0	4.0	6.0

https://github.com/bmaillou/BigDataPython/blob/master/oneX_manyY.csv

```
In [41]: manyY=pd.read_csv('oneX_manyY.csv')
manyY
```

```
Out[41]:
```

	x_values	first_y_values	second_y_values	third_y_values	fourth_y_values
0	1	1	10	4	6
1	2	2	9	4	6
2	3	3	8	4	6
3	4	4	7	4	6
4	5	5	6	4	6
5	6	6	5	4	6
6	7	7	4	4	6
7	8	8	3	4	6
8	9	9	2	4	6
9	10	10	1	4	6

To show you how data sets/types are related we could strip off the column titles so it becomes a 2d array for numpy

```
In [42]: manyY=manyY.values
```

```
In [43]: manyY
```

```
Out[43]: array([[ 1,  1, 10,  4,  6],
                [ 2,  2,  9,  4,  6],
                [ 3,  3,  8,  4,  6],
                [ 4,  4,  7,  4,  6],
                [ 5,  5,  6,  4,  6],
                [ 6,  6,  5,  4,  6],
                [ 7,  7,  4,  4,  6],
                [ 8,  8,  3,  4,  6],
                [ 9,  9,  2,  4,  6],
                [10, 10,  1,  4,  6]])
```

Now you can do all of your array nomenclature. For example lets look at the first column

```
In [44]: manyY[:,0]
```

```
Out[44]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

now we can plot the data. For plotting you just keep on listing the x and y pairs.

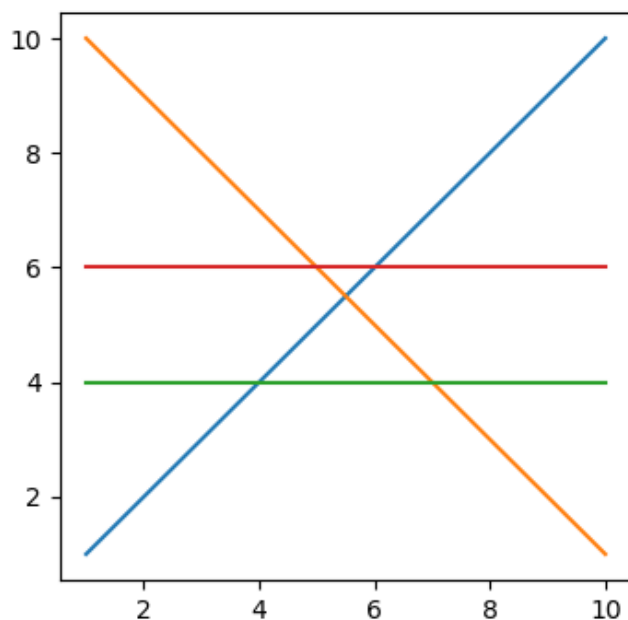
So plot

- column 0 versus column 1, X Value versus First Y Value
- column 0 versus column 2, X Value versus Second Y Value
- column 0 versus column 3, X Value versus Third Y Value
- column 0 versus column 4, X Value versus Fourth Y Value

```
In [45]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
ax.plot(manyY[:,0],manyY[:,1])
ax.plot(manyY[:,0],manyY[:,2])
ax.plot(manyY[:,0],manyY[:,3])
ax.plot(manyY[:,0],manyY[:,4])
```

Out[45]: [

Figure

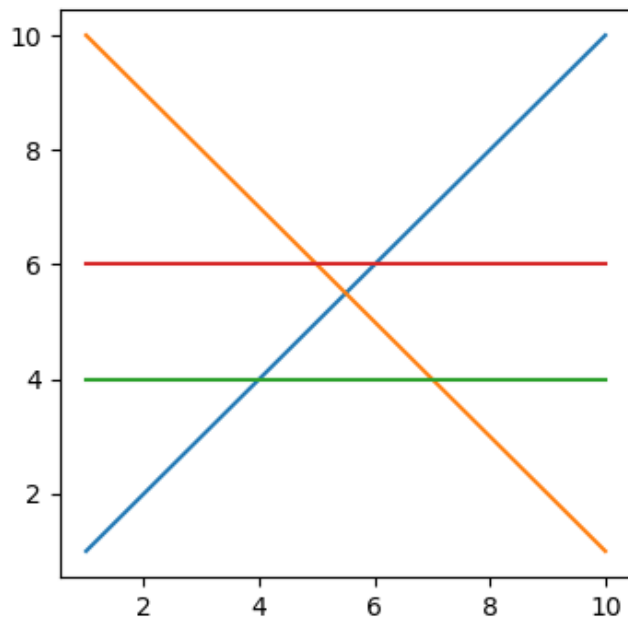


or you can do it in one call to ax.plot by listing x and y pairs but I find this hard to follow

```
In [46]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
ax.plot(manyY[:,0],manyY[:,1]
        ,manyY[:,0],manyY[0:,2]
        ,manyY[:,0],manyY[:,3]
        ,manyY[:,0],manyY[:,4])
```

Out[46]: [<matplotlib.lines.Line2D at 0x147e15090>,
<matplotlib.lines.Line2D at 0x147e24810>,
<matplotlib.lines.Line2D at 0x147de7a10>]

Figure

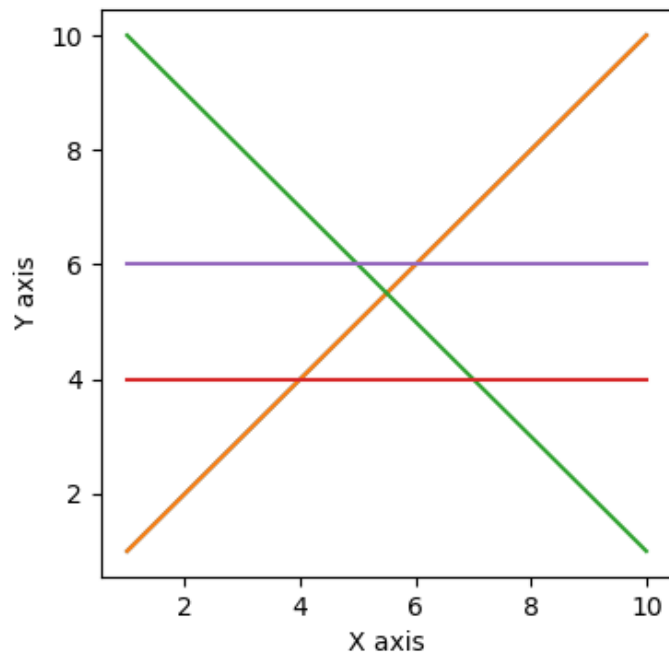


If you wanted to get fancy we could program a for loop to loop over the columns and plot them

```
In [47]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
for i in np.arange(5):
    ax.plot(manyY[:,0],manyY[:,i])
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
```

```
Out[47]: Text(0, 0.5, 'Y axis')
```


Figure

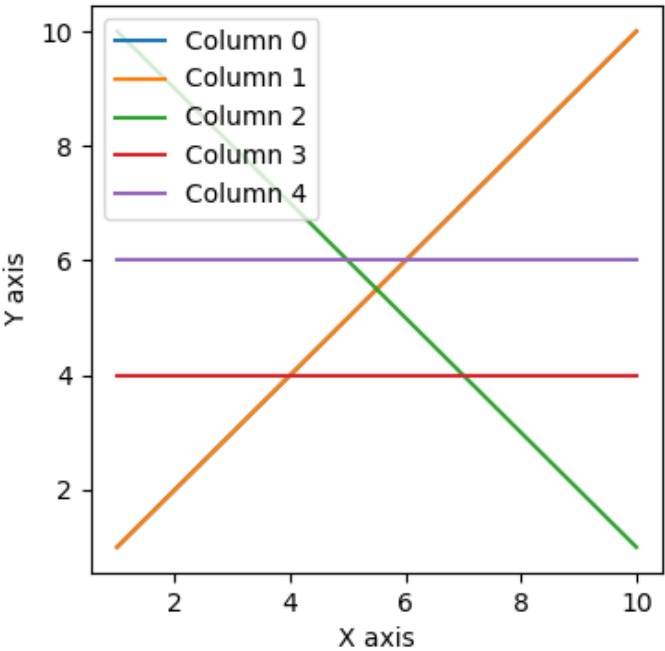


And you can add a legend

```
In [48]: fig,ax=plt.subplots()
fig.set_size_inches(4,4)
for i in np.arange(manyY.shape[1]):
    labeltext='Column '+str(i)
    ax.plot(manyY[:,0],manyY[:,i],label=labeltext)
ax.legend(loc='best')
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
```

```
Out[48]: Text(0, 0.5, 'Y axis')
```

Figure



Compare a 2d array to a Pandas Dataframe

But now lets do it in pandas and see how it compares

```
In [49]: df_manyY=pd.read_csv('oneX_manyY.csv')
df_manyY
```

Out[49]:

	x_values	first_y_values	second_y_values	third_y_values	fourth_y_values
0	1	1	10	4	6
1	2	2	9	4	6
2	3	3	8	4	6
3	4	4	7	4	6
4	5	5	6	4	6
5	6	6	5	4	6
6	7	7	4	4	6
7	8	8	3	4	6
8	9	9	2	4	6
9	10	10	1	4	6

Pandas is like an upgraded numpy array with column names.

This is going to make keeping track of data much nicer

using pandas we can use the column names

```
In [50]: df_manyY.columns
```

```
Out[50]: Index(['x_values', 'first_y_values', 'second_y_values', 'third_y_values',  
              'fourth_y_values'],  
              dtype='object')
```

```
In [51]: df_manyY['x_values']
```

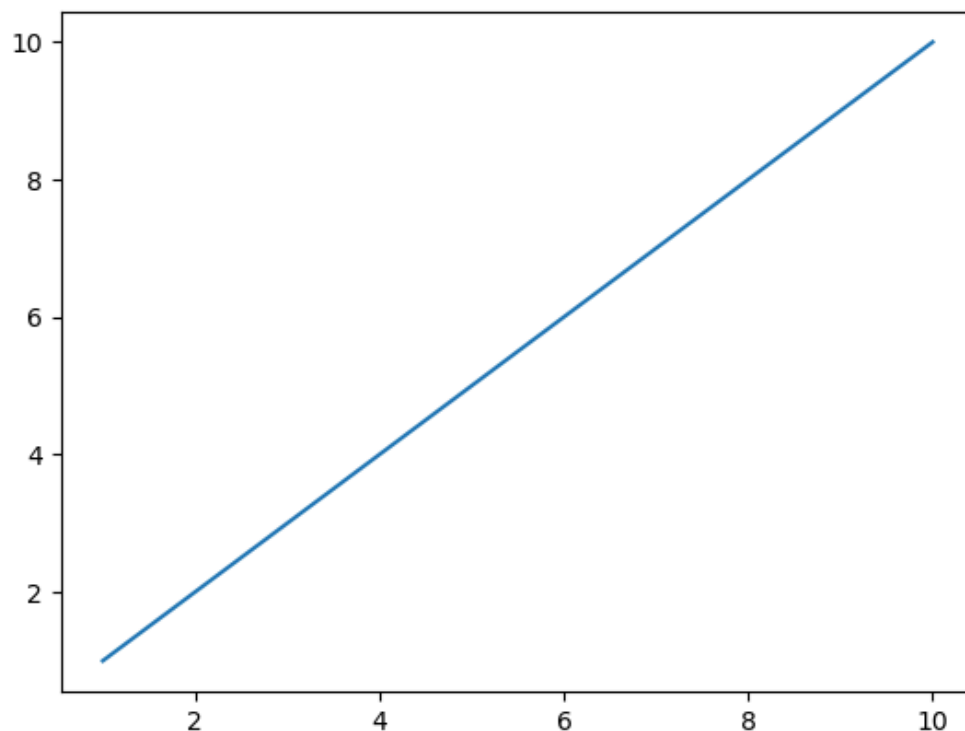
```
Out[51]: 0    1  
         1    2  
         2    3  
         3    4  
         4    5  
         5    6  
         6    7  
         7    8  
         8    9  
         9   10  
Name: x_values, dtype: int64
```

making the plot in pandas

```
In [52]: fig,ax=plt.subplots()  
         ax.plot(df_manyY['x_values'],df_manyY['first_y_values'])
```

```
Out[52]: [matplotlib.lines.Line2D at 0x147f8cc10>]
```

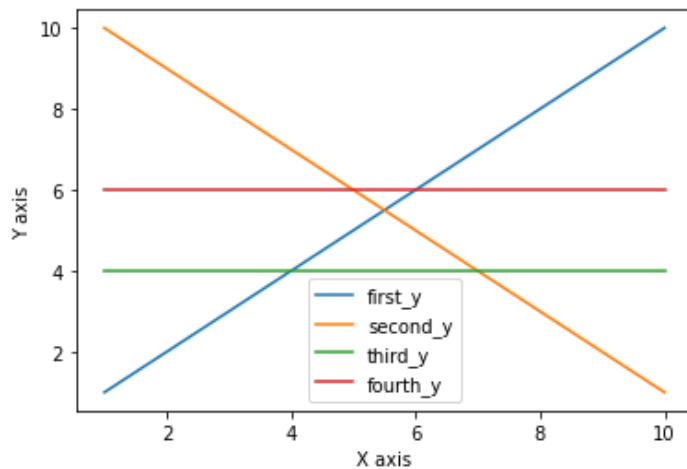
Figure



Can you add the other columns and make a legend? Using Pandas?

In [86]:

Out[86]: Text(0, 0.5, 'Y axis')



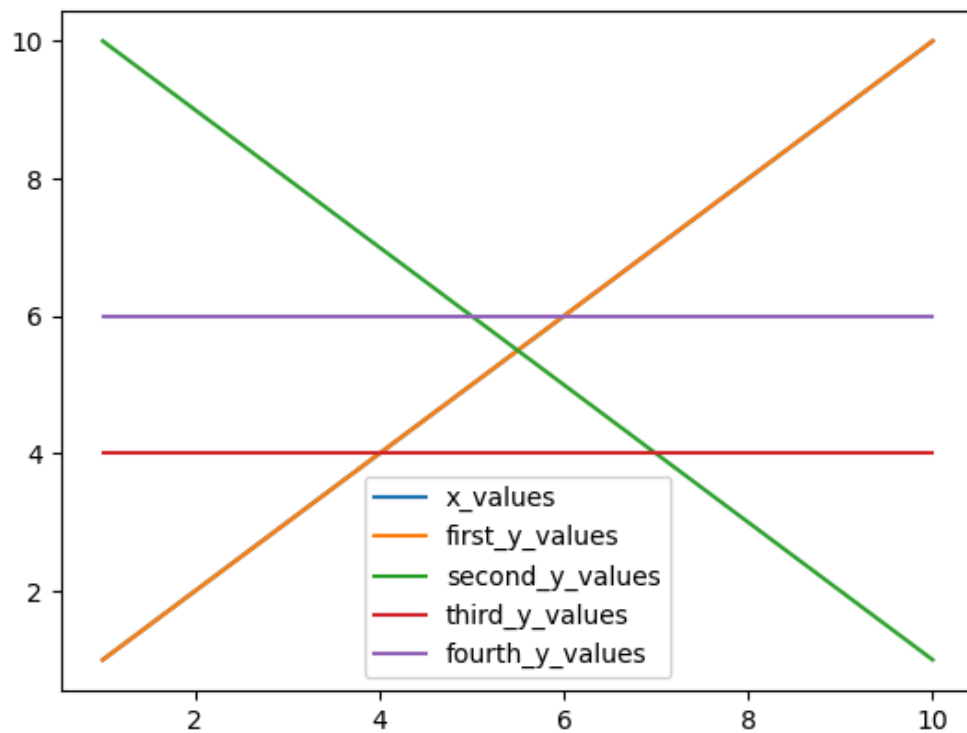
Pandas does some things to make your life easy. You can for loop over the columns. So the for loop returns the column name to col and you can pass that to ax.plot. We are going to be doing a lot more of this the next few weeks. So this is a sneak peak.

In [53]: fig,ax=plt.subplots()

```
for col in df_manyY:  
    ax.plot(df_manyY['x_values'],df_manyY[col],label=col)  
  
ax.legend()
```

Out[53]: <matplotlib.legend.Legend at 0x147fb1dd0>

Figure



But this plots the first column versus itself. So we just need to only call from after the first column. We will learn how to do this next time. But here it is

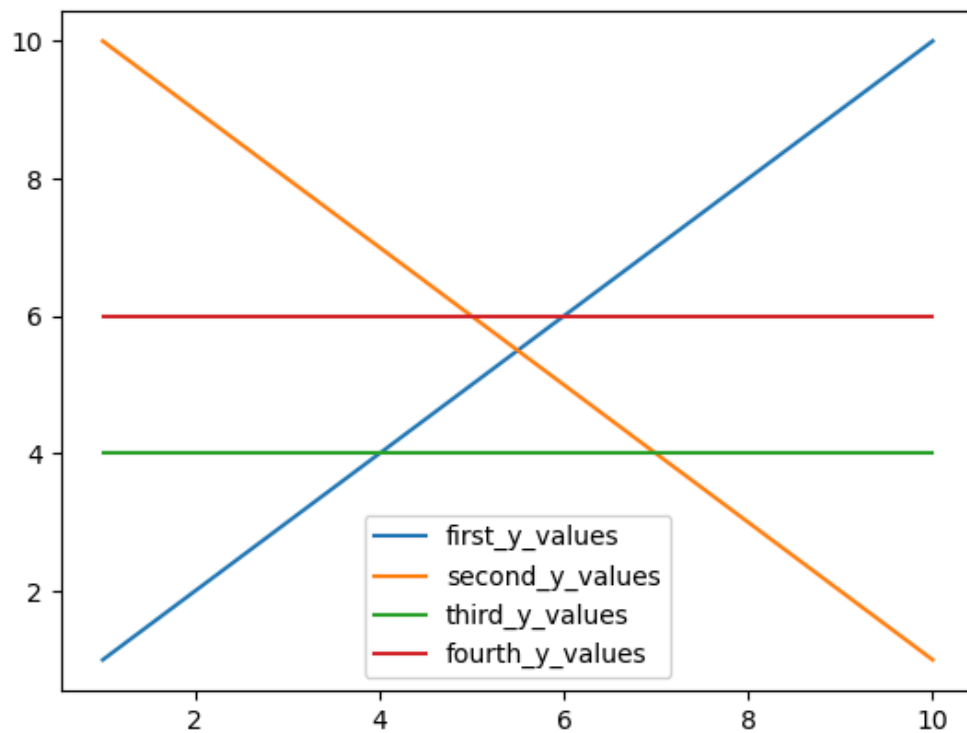
```
In [54]: fig,ax=plt.subplots()

for col in df_manyY.iloc[:,1:]:
    ax.plot(df_manyY['x_values'],df_manyY[col],label=col)

ax.legend()
```

```
Out[54]: <matplotlib.legend.Legend at 0x147f80210>
```

Figure



Mystery file

The file `mystery.csv` contains data in columns. Use what you know and plot the data! The first column is x values. The others are y values

In [143...]

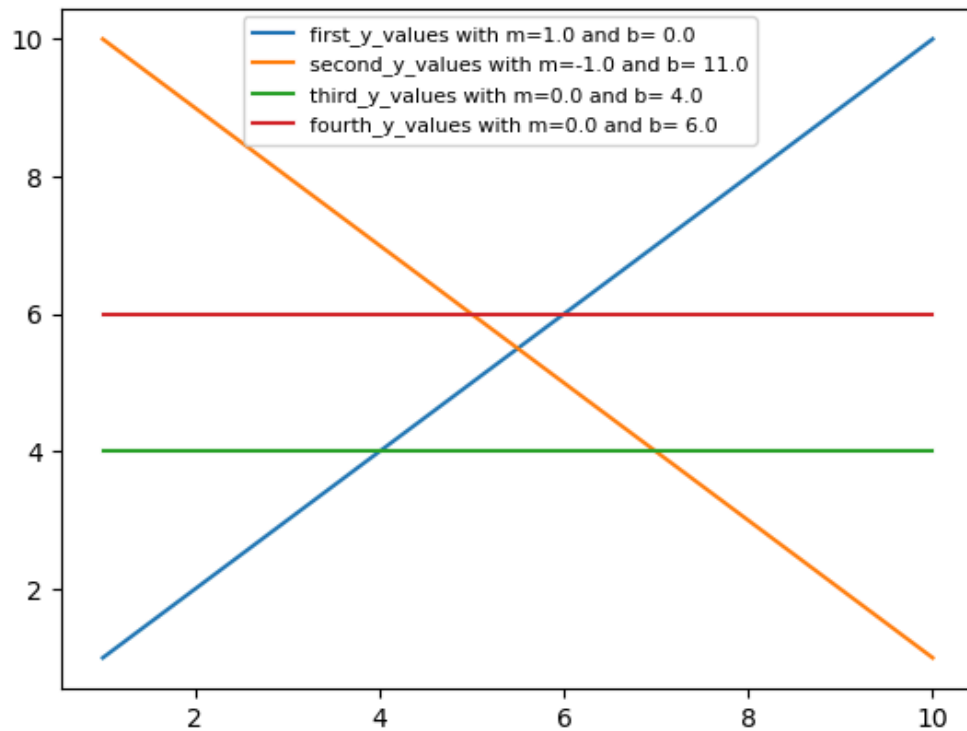
Bonus.

If you got through this quickly see if you can go back to `oneX_manyY.csv` and get the equations for each line. You could do this in a for loop and adding each equation for a line to the legend...

In [74]:

Out[74]: `<matplotlib.legend.Legend at 0x13342e6d0>`

Figure



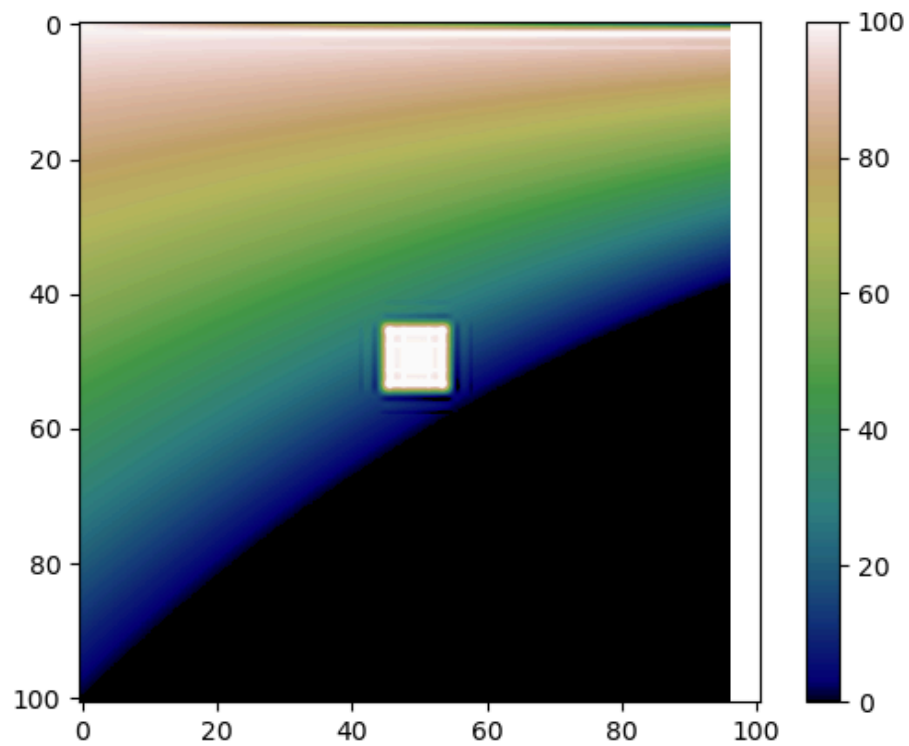
Answers

My own colorbar

```
In [60]: fig,ax=plt.subplots()  
cax=ax.imshow(twoD,cmap='gist_earth',interpolation='bessel')  
fig.colorbar(cax)
```

```
Out[60]: <matplotlib.colorbar.Colorbar at 0x132d5fbe0>
```

Figure

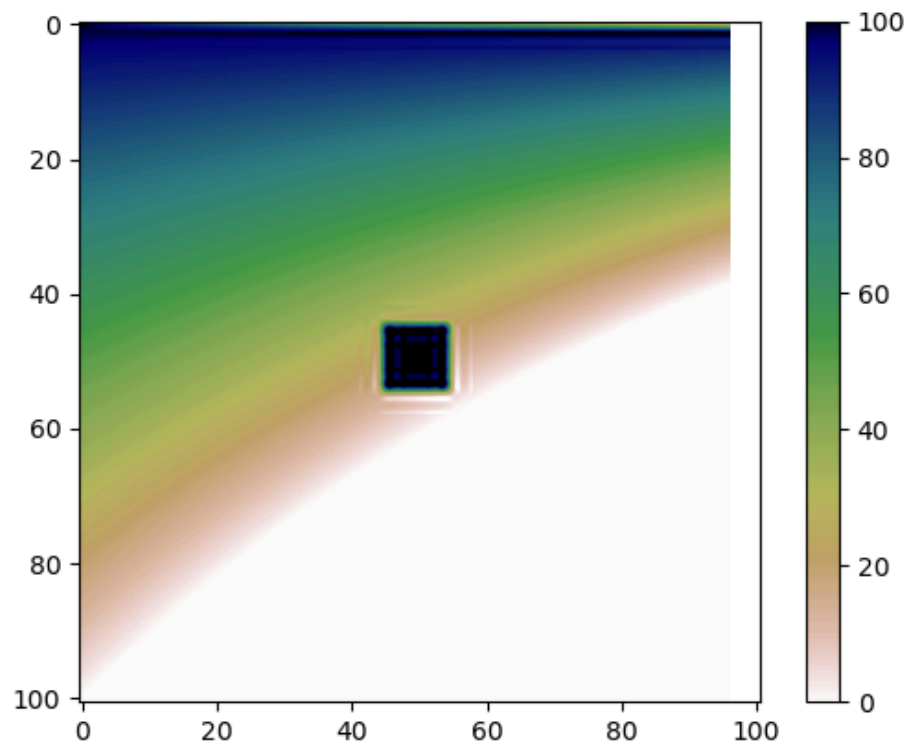


now reversed

```
In [61]: fig,ax=plt.subplots()  
cax=ax.imshow(twoD,cmap='gist_earth_r',interpolation='bessel')  
fig.colorbar(cax)
```

```
Out[61]: <matplotlib.colorbar.Colorbar at 0x132de3820>
```


Figure



Brian result

```
In [62]: Brian=pd.read_csv('Brian.csv',header=None)
         Brian=Brian.values
         fig,ax=plt.subplots()
         ax.imshow(Brian,cmap='gnuplot',interpolation='none')
```

```
Out[62]: <matplotlib.image.AxesImage at 0x132d2e970>
```

Figure

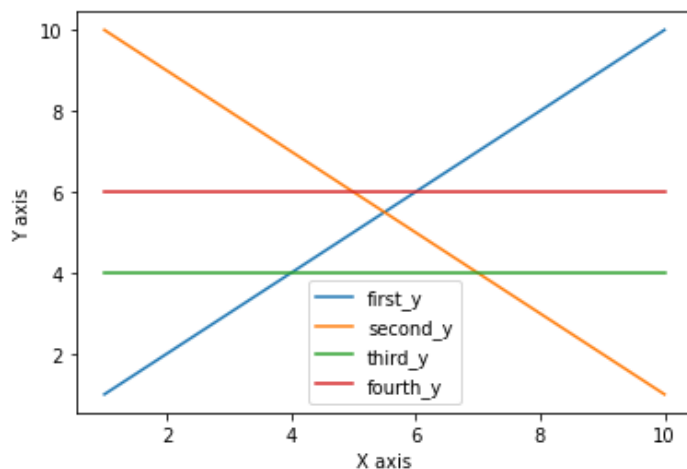


```
In [99]: fig,ax=plt.subplots()

ax.plot(df_manyY['x_values'],df_manyY['first_y_values'],label='first_y')
ax.plot(df_manyY['x_values'],df_manyY['second_y_values'],label='second_y')
ax.plot(df_manyY['x_values'],df_manyY['third_y_values'],label='third_y')
ax.plot(df_manyY['x_values'],df_manyY['fourth_y_values'],label='fourth_y')

ax.legend(loc='best')
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
```

Out[99]: Text(0, 0.5, 'Y axis')



Mystery File

```
In [65]: df_mystery=pd.read_csv('mystery.csv')
df_mystery.columns
```

```
Out[65]: Index(['Column_0', 'Column_1', 'Column_2'], dtype='object')
```

```
In [66]: df_mystery=pd.read_csv('mystery.csv')
fig,ax=plt.subplots()

ax.plot(df_mystery['Column_0'],df_mystery['Column_1'],label='column 1')
ax.plot(df_mystery['Column_0'],df_mystery['Column_2'],label='column 2')

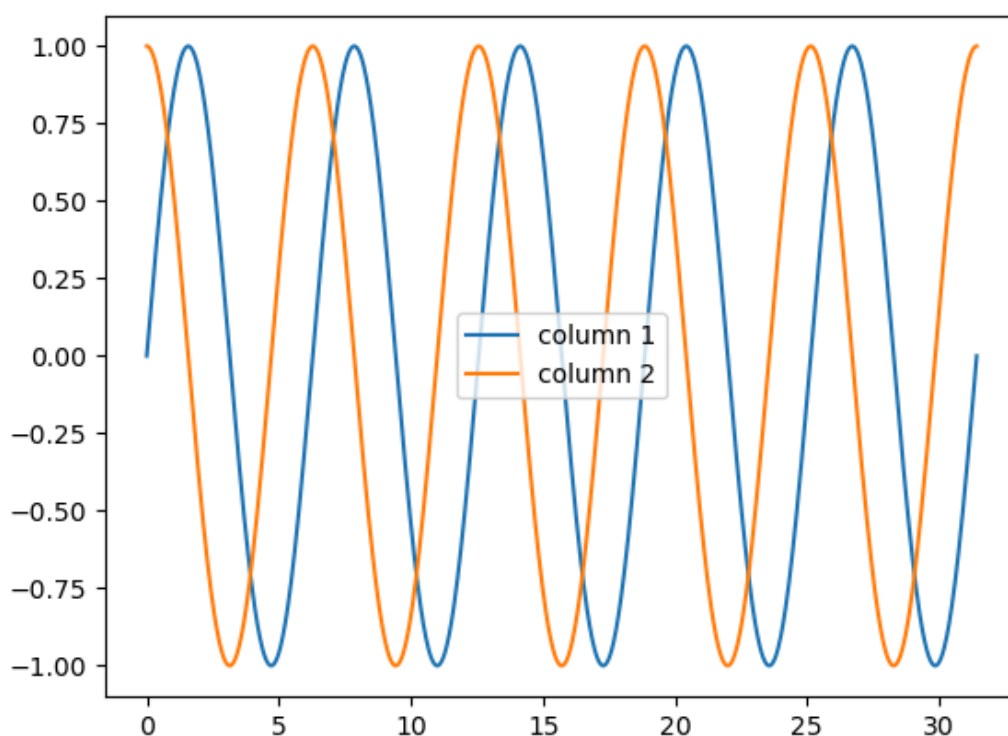
ax.legend()
```

/var/folders/dw/0b9k7c_s7nj6fspm6n2gbk40000gp/T/ipykernel_73564/3091305477.py:2: RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot interface (`matplotlib.pyplot.figure`) are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam `figure.max\_open\_warning`). Consider using `matplotlib.pyplot.close()`.

```
fig,ax=plt.subplots()
```

```
Out[66]: <matplotlib.legend.Legend at 0x133054cd0>
```

Figure



Using a for loop to get all the equations for the lines. Use linregress

```
In [73]: df_manyY=pd.read_csv('oneX_manyY.csv')

fig,ax=plt.subplots()

for col in df_manyY.iloc[:,1:]:
```

```

x=df_manyY['x_values']
y=df_manyY[col]

slope, intercept, r_value,p_value,stderr = stats.linregress(x,y)
label='{} with m={} and b = {}'.format(col,slope,intercept)

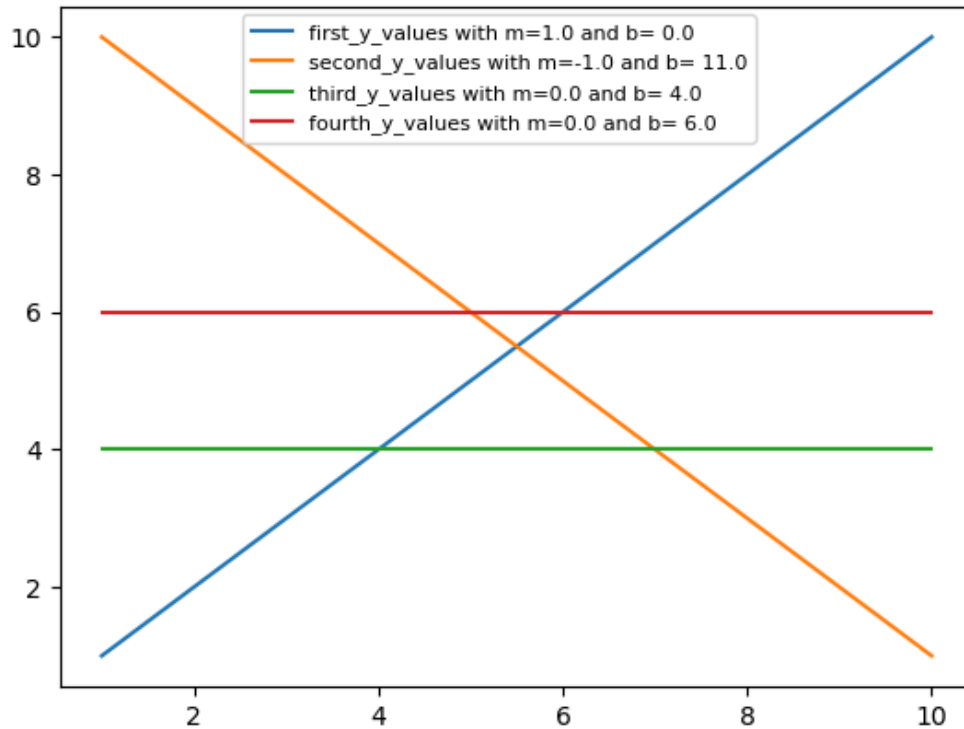
ax.plot(df_manyY['x_values'],df_manyY[col],label=label)

ax.legend(loc=(.2,.8),fontsize=8)

```

Out[73]: <matplotlib.legend.Legend at 0x133390430>

Figure



Code I copied from the web to show all the colormaps

```

In [ ]: """
Reference for colormaps included with Matplotlib.

This reference example shows all colormaps included with Matplotlib. Note that
any colormap listed here can be reversed by appending "_r" (e.g., "pink_r").
These colormaps are divided into the following categories:

Sequential:
    These colormaps are approximately monochromatic colormaps varying smoothly
    between two color tones—usually from low saturation (e.g. white) to high
    saturation (e.g. a bright blue). Sequential colormaps are ideal for
    representing most scientific data since they show a clear progression from
    low-to-high values.

Diverging:
    These colormaps have a median value (usually light in color) and vary

```

smoothly to two different color tones at high and low values. Diverging colormaps are ideal when your data has a median value that is significant (e.g. 0, such that positive and negative values are represented by different colors of the colormap).

Qualitative:

These colormaps vary rapidly in color. Qualitative colormaps are useful for choosing a set of discrete colors. For example::

```
color_list = plt.cm.Set3(np.linspace(0, 1, 12))
```

gives a list of RGB colors that are good for plotting a series of lines on a dark background.

Miscellaneous:

Colormaps that don't fit into the categories above.

```
"""
```

```
import numpy as np
import matplotlib.pyplot as plt

cmaps = [('Sequential',      ['Blues', 'BuGn', 'BuPu',
                              'GnBu', 'Greens', 'Greys', 'Oranges', 'OrRd',
                              'PuBu', 'PuBuGn', 'PuRd', 'Purples', 'RdPu',
                              'Reds', 'YlGn', 'YlGnBu', 'YlOrBr', 'YlOrRd']),
          ('Sequential (2)', ['afmhot', 'autumn', 'bone', 'cool', 'copper',
                              'gist_heat', 'gray', 'hot', 'pink',
                              'spring', 'summer', 'winter']),
          ('Diverging',      ['BrBG', 'bwr', 'coolwarm', 'PiYG', 'PRGn', 'PuOr',
                              'RdBu', 'RdGy', 'RdYlBu', 'RdYlGn', 'Spectral',
                              'seismic']),
          ('Qualitative',    ['Accent', 'Dark2', 'Paired', 'Pastel1',
                              'Pastel2', 'Set1', 'Set2', 'Set3']),
          ('Miscellaneous',  ['gist_earth', 'terrain', 'ocean', 'gist_stern',
                              'brg', 'CMRmap', 'cubehelix',
                              'gnuplot', 'gnuplot2', 'gist_ncar',
                              'nipy_spectral', 'jet', 'rainbow',
                              'gist_rainbow', 'hsv', 'flag', 'prism'])]

nrows = max(len(cmap_list) for cmap_category, cmap_list in cmaps)
gradient = np.linspace(0, 1, 256)
gradient = np.vstack((gradient, gradient))

def plot_color_gradients(cmap_category, cmap_list):
    fig, axes = plt.subplots(nrows=nrows)
    fig.subplots_adjust(top=0.95, bottom=0.01, left=0.2, right=0.99)
    axes[0].set_title(cmap_category + ' colormaps', fontsize=14)

    for ax, name in zip(axes, cmap_list):
        ax.imshow(gradient, aspect='auto', cmap=plt.get_cmap(name))
        pos = list(ax.get_position().bounds)
        x_text = pos[0] - 0.01
        y_text = pos[1] + pos[3]/2.
        fig.text(x_text, y_text, name, va='center', ha='right', fontsize=10)

    # Turn off *all* ticks & spines, not just the ones with colormaps.
    for ax in axes:
        ax.set_axis_off()
```

```
for cmap_category, cmap_list in cmaps:  
    plot_color_gradients(cmap_category, cmap_list)  
  
plt.show()
```

In []:

In []:

In []: